Score-Based Diffusion Models for 2D and 3D ϕ^4 Lattice Field Theory

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Outline

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Critical Points from Wolff + FA-HMC

Score-Based Diffusion Model

Training Diagnostics

2D Propagator Results

3D Results

Summary

Critical Points from Wolff + FA-HMC

Lattice ϕ^4 Action

The scalar ϕ^4 theory on a D-dimensional L^D lattice with periodic BC:

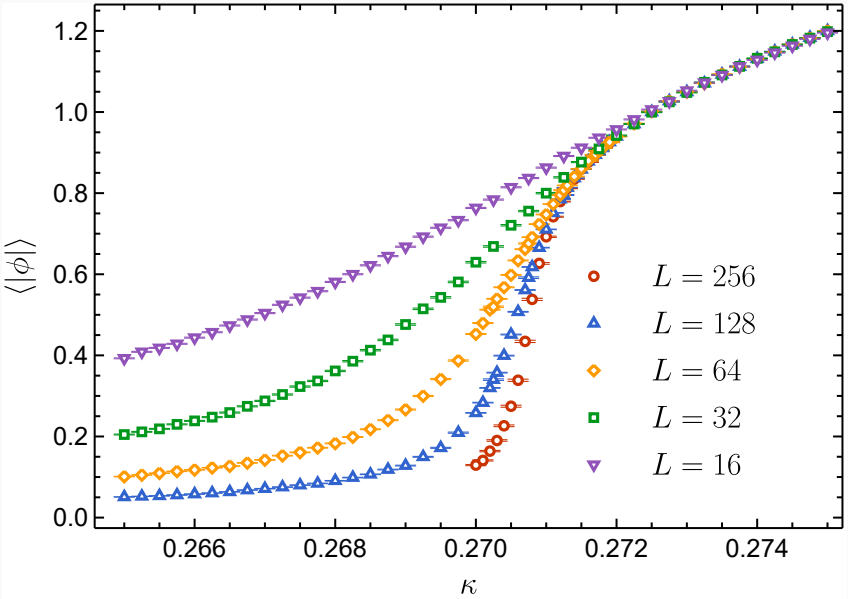
$$S[\phi] = \sum_x \Big[-2\kappa \sum_{\mu=1}^{D} \phi_x \, \phi_{x+\hat{\mu}} + \phi_x^2 + \lambda \big(\phi_x^2 - 1\big)^2 \Big] \tag{1}$$

- κ: hopping parameter, λ: quartic self-coupling
- $\mathbb{Z}_2$-breaking phase transition at $\kappa_c(\lambda)$
- **2D**: $\lambda = 0.022$, **3D**: $\lambda = 0.9$

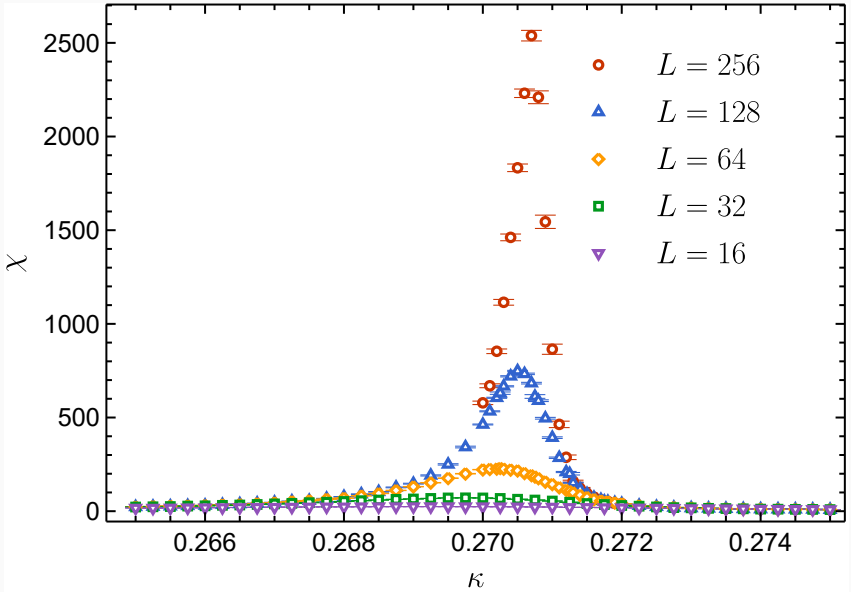
Sampling algorithm: Wolff cluster flips + Fourier-accelerated HMC

- FA-HMC: momentum-dependent mass $M(\hat{p}) = \hat{p}^2 + m_{\text{eff}}^2$ via FFT
- Wolff cluster: flips $\phi_x \to -\phi_x$ to decorrelate magnetisation

2D: Order Parameter and Susceptibility

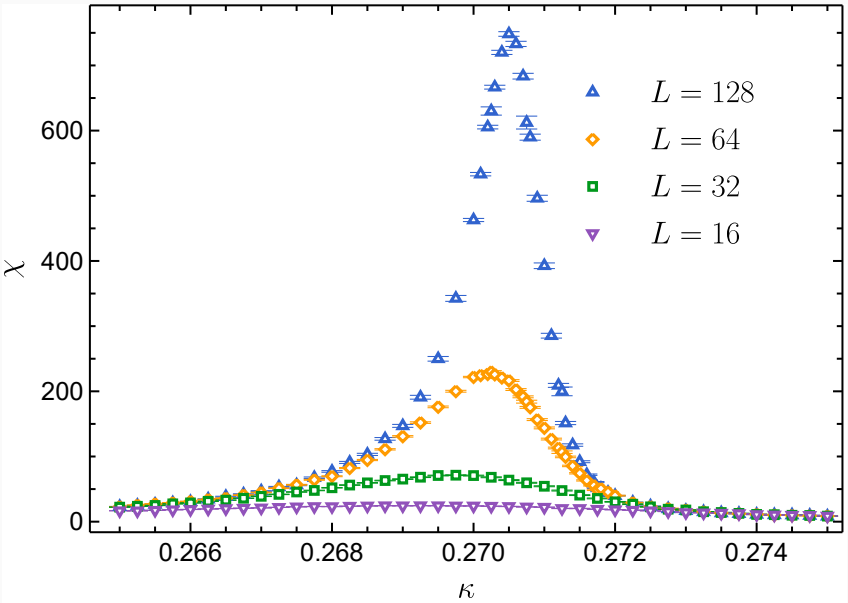


Order parameter $\langle|\bar{\phi}|\rangle$ vs. κ for $L = 16$–256. Transition sharpens with L.

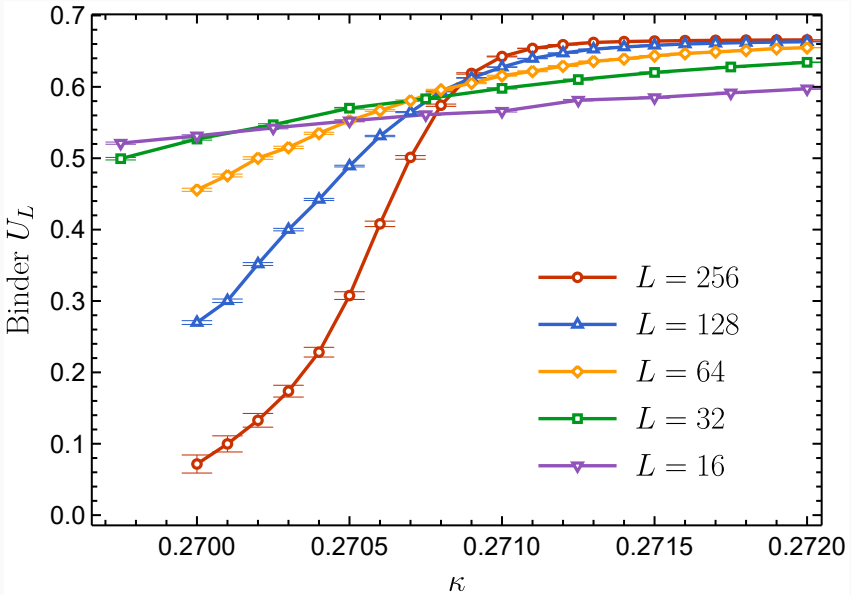


Susceptibility χ vs. κ. Peak grows $\sim L^{\gamma/\nu}$ ($\gamma/\nu = 7/4$, 2D Ising).

2D: Susceptibility and Binder Cumulant

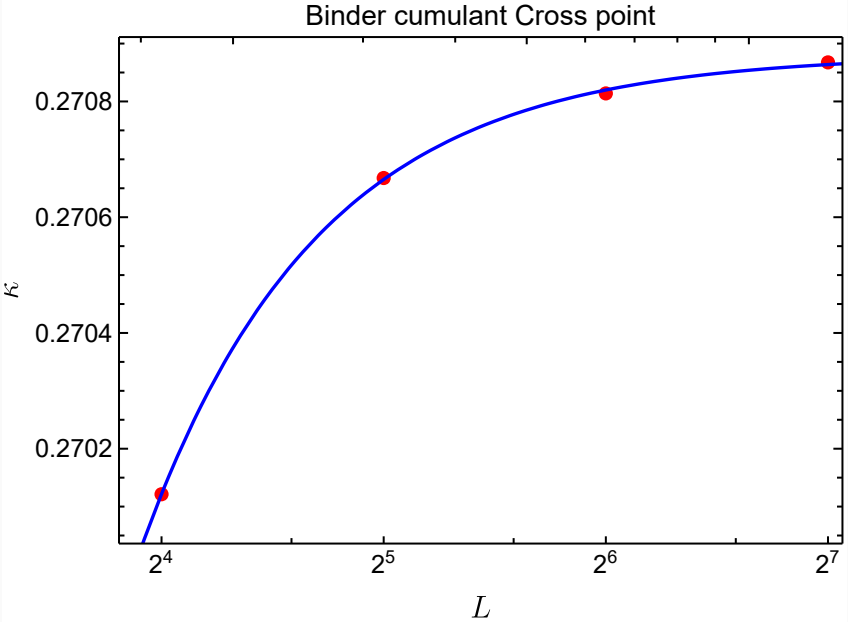


Susceptibility χ (alternative representation).



Binder cumulant U_L vs. κ for $L = 16$–256. Curves cross at κ_c.

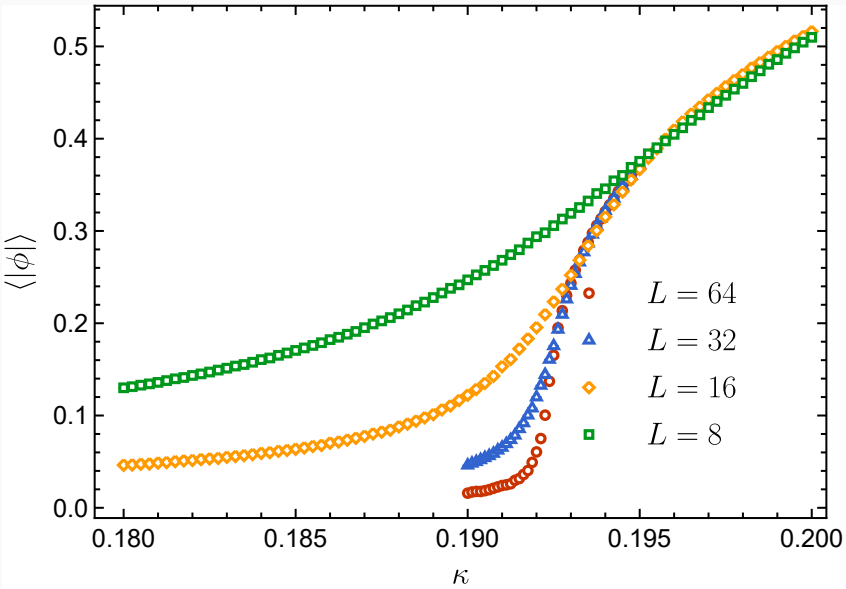
2D Critical Point: $\kappa_c \approx 0.27088$



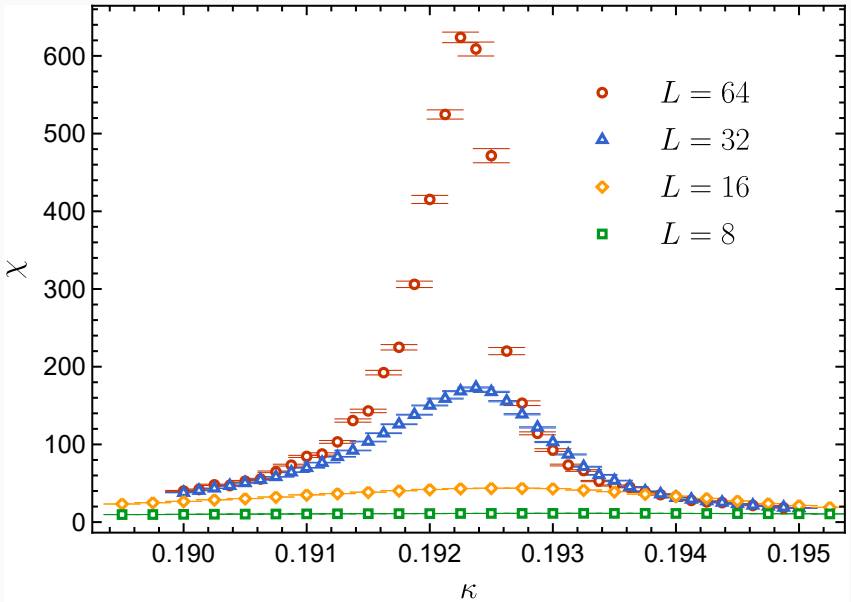
Binder crossing analysis $\Rightarrow$
$\kappa_c(\lambda = 0.022) \approx 0.27088$.

- 2D Ising universality class
- $\lambda = 0.022$
- Three κ values for DM study:
  - 0.26 (symmetric)
  - $0.2705 \approx \kappa_c$ (critical)
  - 0.28 (broken)

3D: Order Parameter and Susceptibility

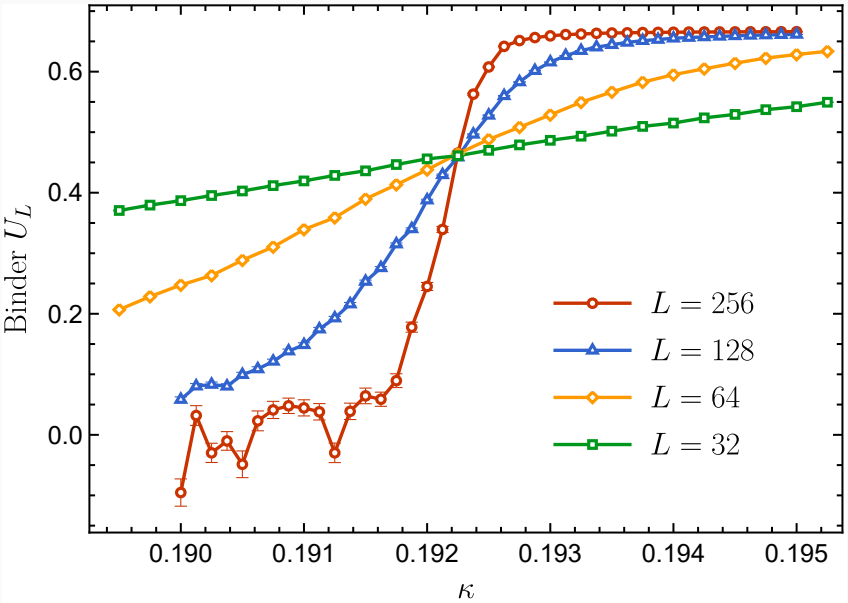


Order parameter $\langle|\bar{\phi}|\rangle$ vs. κ for $L = 8$–64.



Susceptibility χ vs. κ. Peak converges toward κ_c.

3D Critical Point: $\kappa_c \approx 0.19225$



Binder cumulant crossing $\Rightarrow$
$\kappa_c(\lambda = 0.9) \approx 0.19225$.

- 3D Ising universality class ($\nu \approx 0.63$)
- $\lambda = 0.9$
- Three κ values for DM study:
  - 0.18 (symmetric)
  - $0.1923 \approx \kappa_c$ (critical)
  - 0.2 (broken)

Score-Based Diffusion Model

Score-Based Generative Model: Setup

Forward SDE (Variance-Exploding):

$$\mathrm{d}\phi = \sigma^t \, \mathrm{d}W, \quad \sigma(t) = \sqrt{(\sigma^{2t} - 1)/2 \log \sigma} \tag{2}$$

Reverse SDE (sampling):

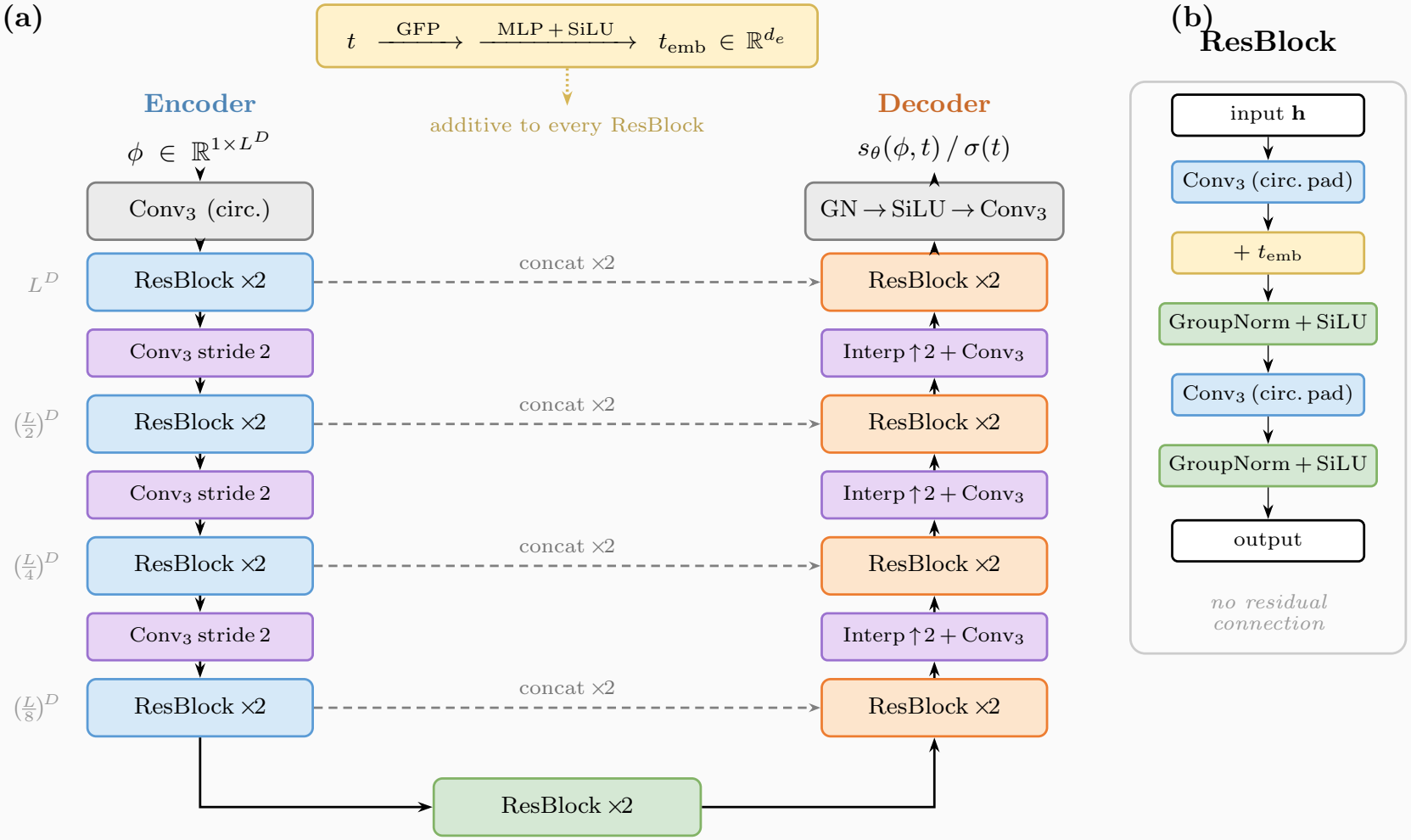
$$\mathrm{d}\phi = -g^2(t) \, \nabla_\phi \log p_t(\phi) \, \mathrm{d}t + g(t) \, \mathrm{d}\bar{W} \tag{3}$$

Score matching loss:

$$\mathcal{L} = \mathbb{E}_{t,\phi_0,z} \left[\left\| s_\theta(\phi_0 + \sigma(t)z, \; t) \cdot \sigma(t) + z \right\|^2 \right] \tag{4}$$

- Train $s_\theta(\phi, t) \approx \nabla_\phi \log p_t(\phi)$
- Sample via Euler–Maruyama reverse integration
- Data normalised to $[-1, 1]$ using global min/max

NCSN++ Architecture



- **Circular padding** on all convolutions ⇒ periodic BC
- Time t encoded via Gaussian Fourier Projection (GFP) + MLP, additive
- 2D: `Conv2d` ($L = 128$); 3D: `Conv3d` ($L = 64$)

Training Diagnostics

MALA Acceptance Rate as Diagnostic

MALA proposal using the score network:

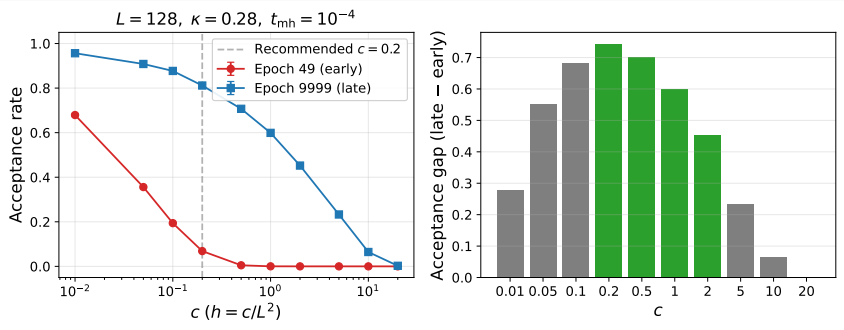
$$\psi = \phi + h\, s_\theta(\phi, t_{\mathrm{mh}}) + \sqrt{2h}\,\eta, \quad \eta \sim \mathcal{N}(0, \mathbb{1}) \tag{5}$$

Accept/reject with exact action $e^{-S(\phi)}$:

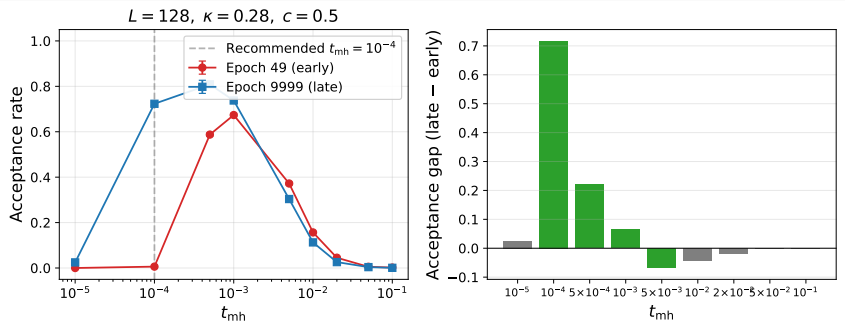
$$\alpha = \min\left\{1,\ \frac{e^{-S(\psi)}\, q(\phi|\psi)}{e^{-S(\phi)}\, q(\psi|\phi)}\right\} \tag{6}$$

- A well-trained score $\Rightarrow$ high acceptance rate
- Step size: $h = c/L^D$ $\quad$ ($c = 0.2$ universal across D)
- Evaluation time: $t_{\mathrm{mh}} = 10^{-4}$ $\quad$ (maximises diagnostic power)
- $N = 1024$ (2D) or $N = 256$ (3D) reference configs, single MH step

2D: Step-Size and t_{mh} Calibration



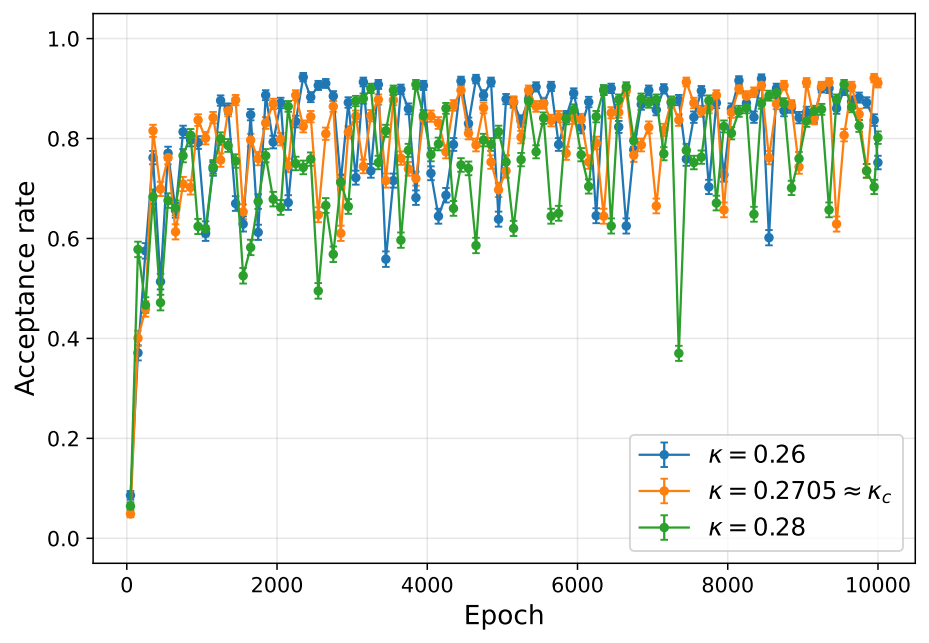
$h = c/L^2$: optimal $c = 0.2$ at $L = 128$.



$t_{\mathrm{mh}} = 10^{-4}$ maximises diagnostic power.

Gap reverses for $t_{\mathrm{mh}} \gtrsim 5 \times 10^{-3}$.

2D: Acceptance Rate vs. Training Epoch



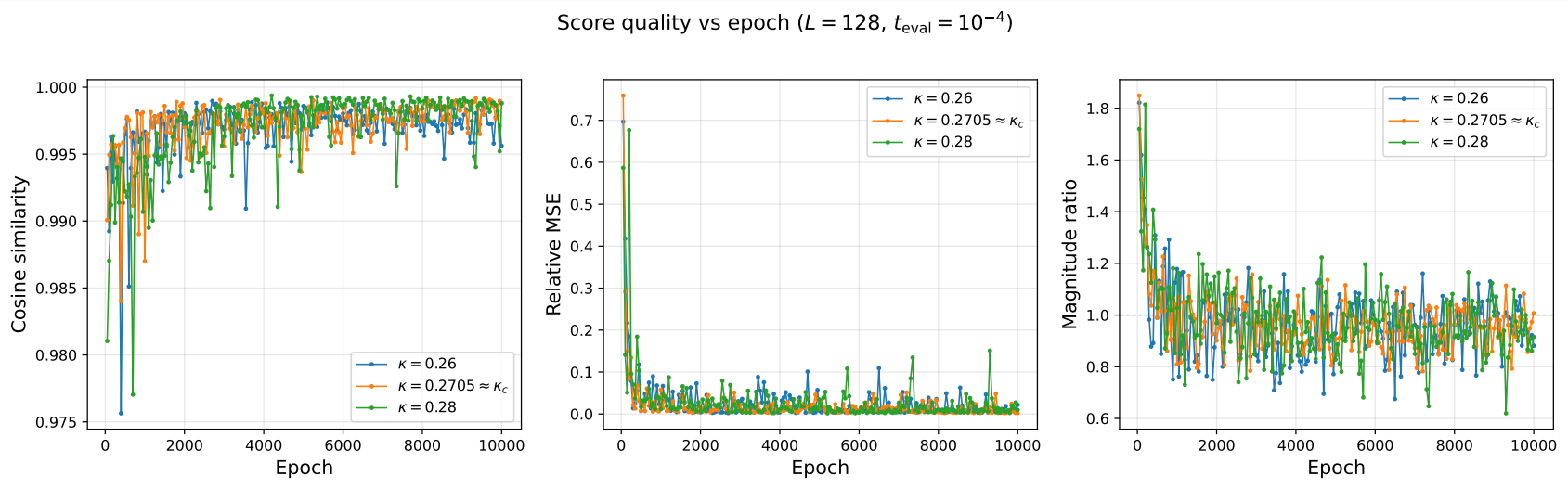
$L = 128$, three κ values. Rises from ~ 0 to 60–90%.

Late training (ep. 3999–4999):

L	$\kappa = 0.26$	0.28
8	0.64	0.63
32	0.68	0.68
64	0.83	0.75
128	0.79	0.75

Acceptance rate increases with L (conservative $h = c/L^2$ scaling).

2D: Direct Score Quality

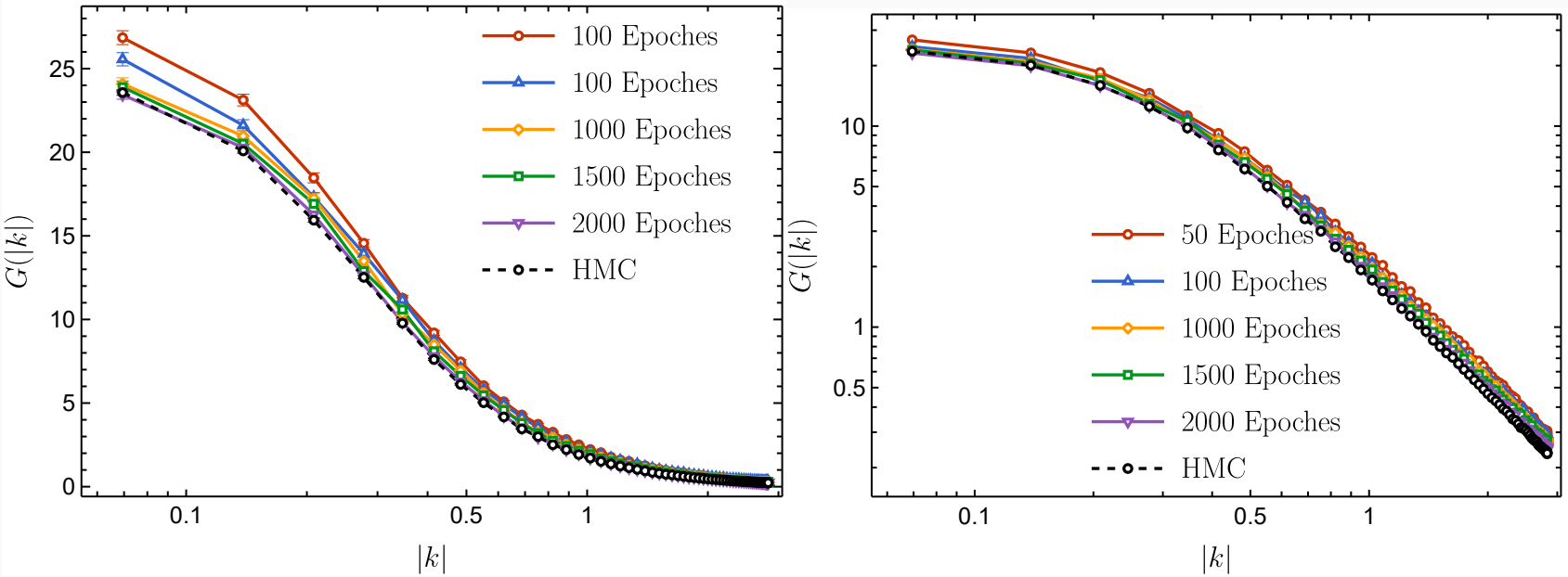


Compare $s_\theta(\phi, t)$ vs. $-\nabla_\phi S$ at $t_{\text{eval}} = 10^{-4}$:

- Cosine similarity: $0.99 \to 0.998$
- Relative MSE: drops $> 10\times$
- Magnitude ratio: $\sim 1.5 \to \sim 0.95$ (converges to 1)

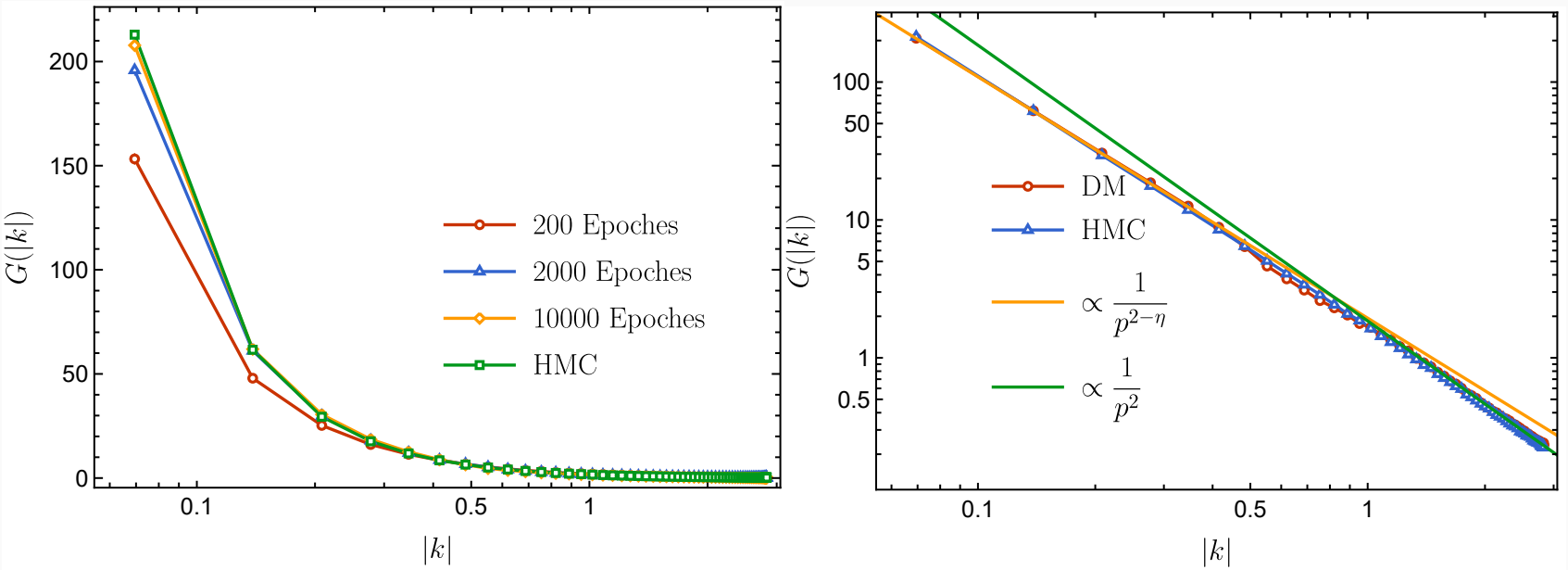
2D Propagator Results

2D Propagator: $\kappa = 0.26$ (Symmetric Phase)



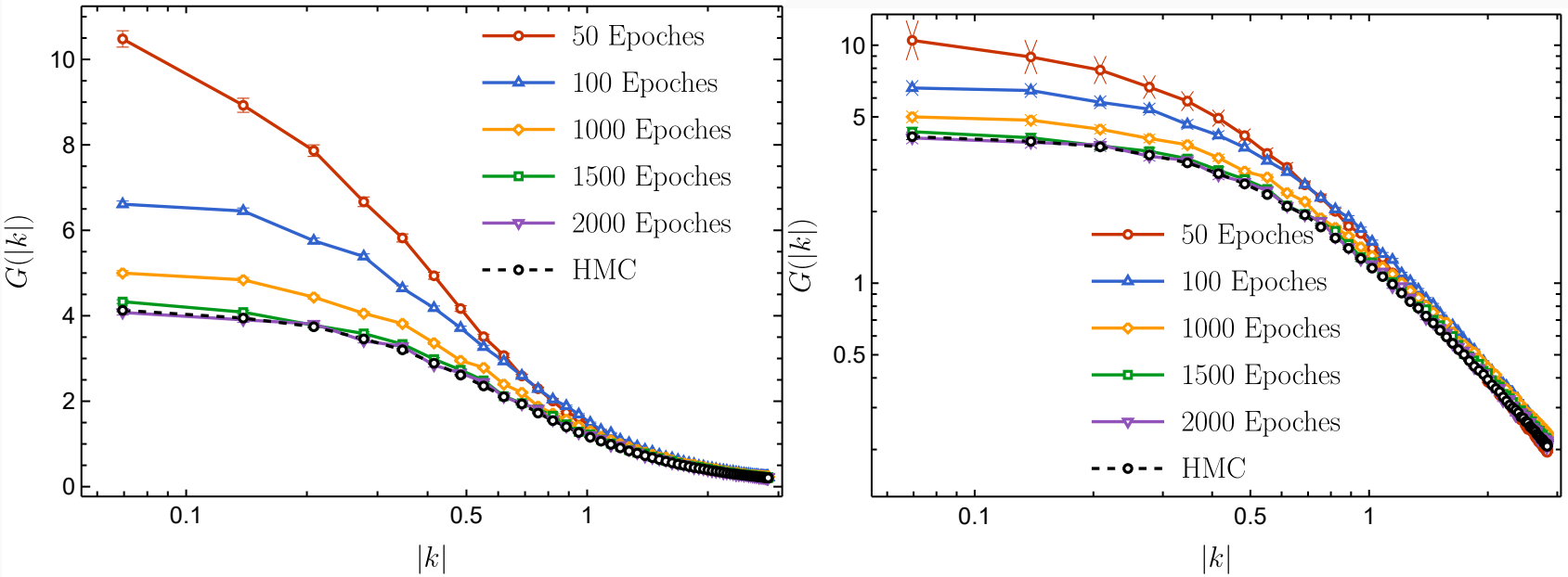
$L = 128$, $\lambda = 0.022$. DM converges to HMC propagator with training.

2D Propagator: $\kappa = 0.2705$ (Near-Critical)



Most demanding: $\xi \to \infty$, power-law tail $G(p) \sim p^{-(2-\eta)}$. DM captures the IR behaviour.

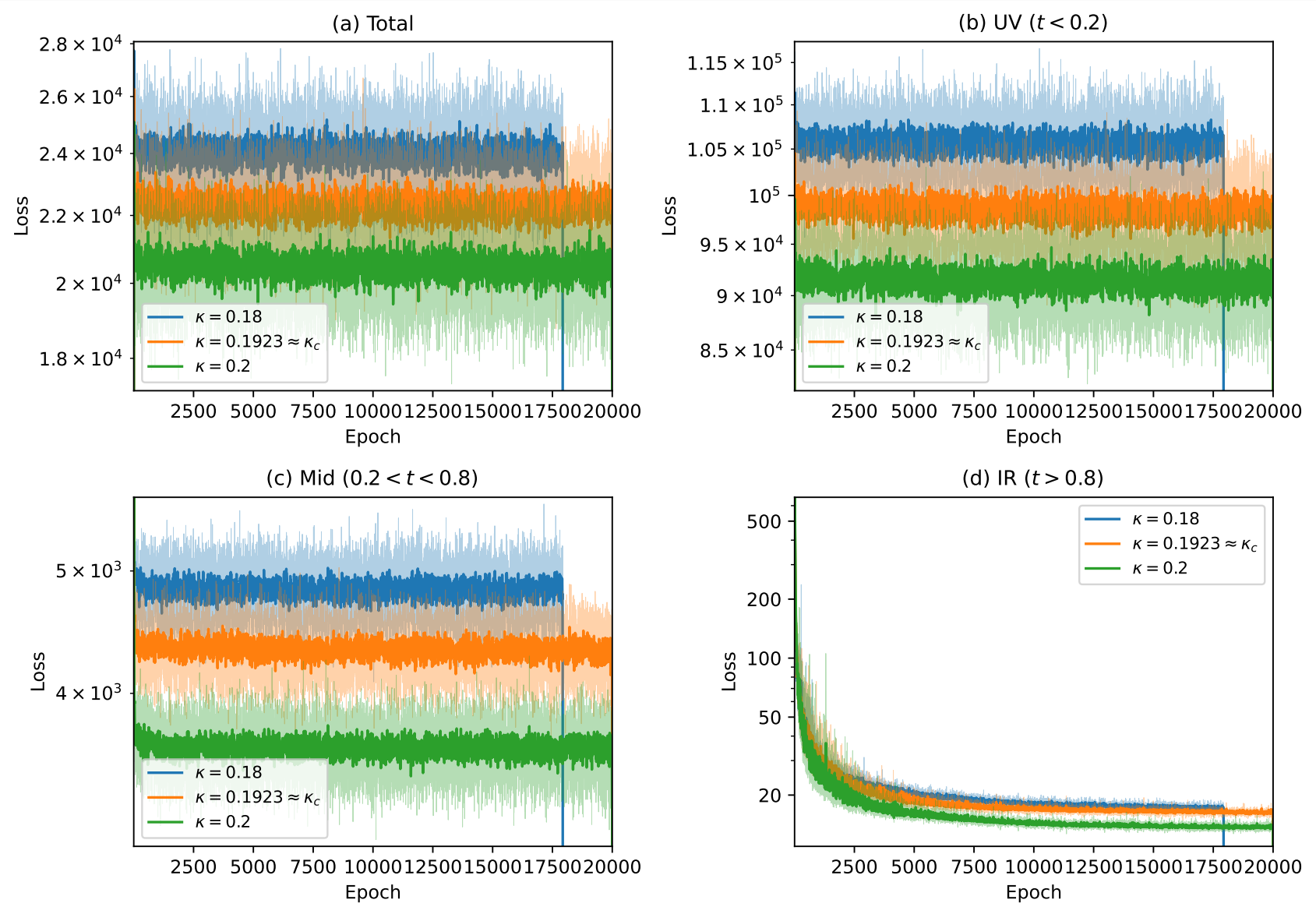
2D Propagator: $\kappa = 0.28$ (Broken Phase)



Ordered phase with mass gap. Faithful reproduction by DM.

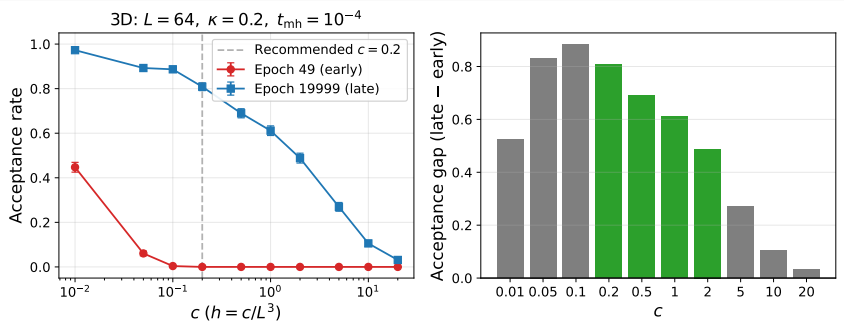
3D Results

3D Training Loss Decomposition



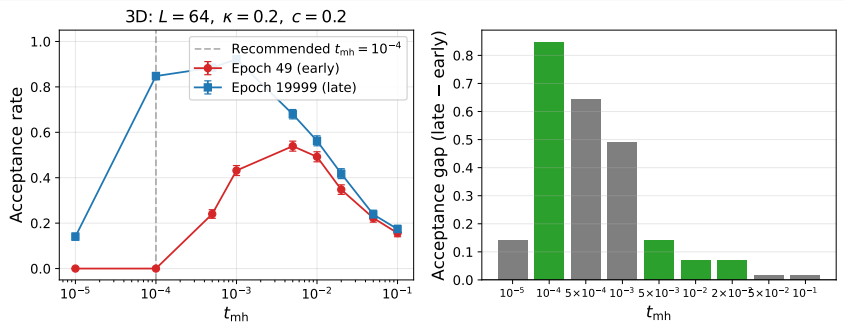
$L = 64$, $\lambda = 0.9$. Loss decomposed by diffusion time t:

3D: Acceptance Rate Calibration



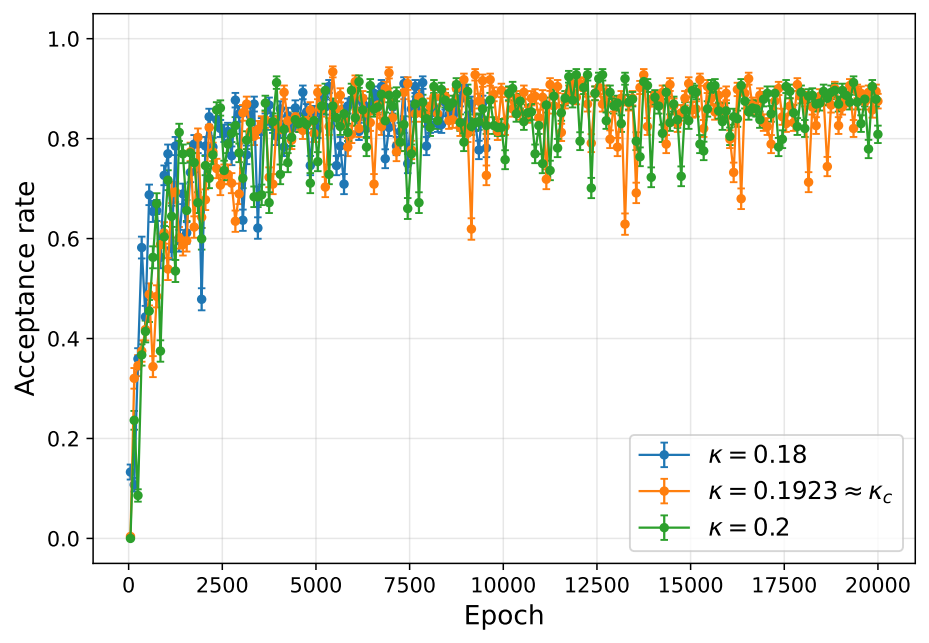
$h = c/L^3$: optimal $c = 0.2$ (same as 2D!).

Validates $h = c/L^D$ scaling.



$t_{mh} = 10^{-4}$ optimal (consistent with 2D).

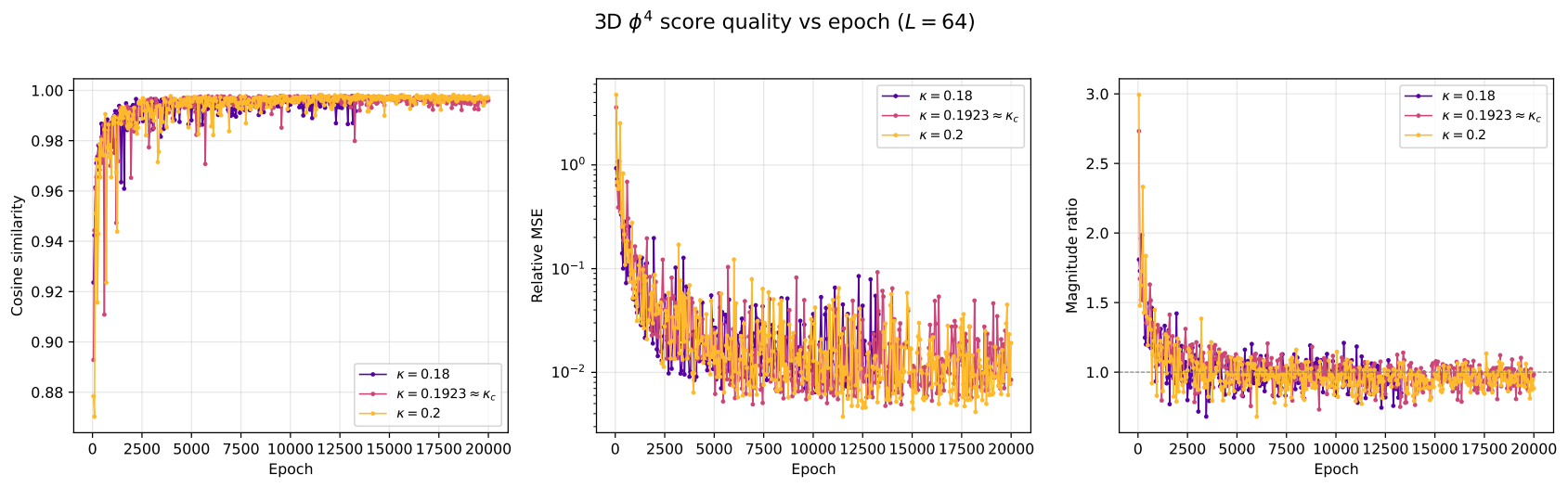
3D: Acceptance Rate vs. Training Epoch



$L = 64$, $\lambda = 0.9$, $c = 0.5$, $h = c/L^3$.

- Acceptance rate reaches 60–90% at late epochs
- Comparable to 2D results
- Confirms $h = c/L^D$ diagnostic across dimensions

3D: Direct Score Quality



$L = 64$, $\lambda = 0.9$, evaluated at $t_{\text{eval}} = 10^{-4}$ with $N = 64$ reference configs:

- Cosine similarity: $0.96 \to 0.995$
- Relative MSE: drops $> 10\times$
- Magnitude ratio: 1.4–1.5 $\to$ 0.97–1.00 (mirrors 2D trends)

Summary

Summary

1. **Critical points** established via Wolff + FA-HMC:
   - 2D: $\kappa_c(\lambda{=}0.022) \approx 0.27088$
   - 3D: $\kappa_c(\lambda{=}0.9) \approx 0.19225$

2. **NCSN++ diffusion model** faithfully reproduces $G(p)$ across all phases (symmetric, critical, broken) for both 2D ($L = 128$) and 3D ($L = 64$)

3. **Diagnostic tools**:
   - MALA acceptance rate with $h = c/L^D$ ($c = 0.2$, universal)
   - Direct score quality: cosine similarity > 0.99, magnitude ratio $\to 1$
   - Loss decomposition (UV/mid/IR) identifies phase-dependent difficulty

4. **Key design**: circular padding enforces periodic BC on all convolutions

Backup

MALA Log-Acceptance Probability

Substituting the proposal $\psi - \phi - h\, s_\theta(\phi) = \sqrt{2h}\,\eta$:

$$\begin{aligned}\log \alpha = \big[S(\phi) - S(\psi)\big] + \tfrac{1}{2}\|\eta\|^2 \\ - \tfrac{1}{4h}\Big\|\sqrt{2h}\,\eta + h\big[s_\theta(\phi) + s_\theta(\psi)\big]\Big\|^2\end{aligned} \tag{7}$$

- Single MH step avoids warm-up bias
- $h = c/L^D$ rather than MALA-optimal $d^{-1/3}$ due to nearest-neighbour correlations
- Gap reverses sign at large t_{mh}: score evaluates $\nabla \log p_t \neq \nabla \log p_0$

2D Acceptance Rate: All Lattice Sizes

Early (epochs 49–249):

L	$\kappa = 0.26$	0.2705	0.28
8	0.152	0.136	0.129
16	0.082	0.228	0.031
32	0.207	0.205	0.297
64	0.288	0.375	0.300
128	0.336	0.325	0.296

Late (epochs 3999–4999):

L	$\kappa = 0.26$	0.2705	0.28
8	0.635	0.715	0.627
16	0.683	0.641	0.693
32	0.680	0.753	0.683
64	0.827	0.818	0.747
128	0.790	0.792	0.749

$c = 0.2$, $t_{\mathrm{mh}} = 10^{-4}$, $N = 1024$, single MH step.

2D Score Quality: All Lattice Sizes

Early (epochs 49–249):

	Mag. ratio		
L	0.26	0.2705	0.28
8	2.10	2.02	1.94
32	1.70	1.65	1.57
128	1.51	1.47	1.46

Late (epochs 3999–4999):

	Mag. ratio		
L	0.26	0.2705	0.28
8	1.28	1.15	1.27
32	1.05	1.01	0.95
128	0.91	0.94	0.95

Cosine similarity > 0.99 for all $L \geq 32$ at late training.

Magnitude ratio converges to ~ 1 for $L \geq 32$.